

Nabd · نبض

Read where the network goes silent.

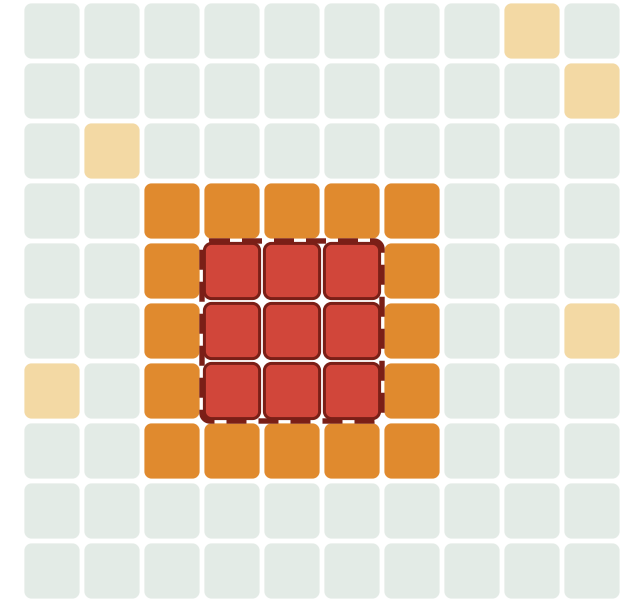
An AI agent that turns coordinated network silence into a live disaster-impact map for command centres — the impact map in the first minute, triage across the first 72 hours.

Theme 6 — Climate Resilience & Environmental Monitoring

Nokia Network-as-Code

Agent layer: LangGraph

Kadir's Team



The first hours decide who is found alive.

53,000+

Kahramanmaraş, Türkiye
earthquakes, February 2023

~3,000

Al Haouz, Morocco
earthquake, September 2023

11,000+

Derna, Libya
dam collapse and flood, September
2023

In each of them, sensors confirmed *something* had happened within seconds.
Which neighbourhoods had lost their people — that map arrived hours late.

2024: Dubai and Oman floods. Same region, a different hazard, the same missing map.

Detection is solved. Impact is not.

SECONDS

Seismometers, satellites and phone-based systems say “**it happened**” almost instantly.

HOURS

“**Which district first?**” is pieced together from fragmented calls, social media and field reports — on a network that is overloaded at exactly that moment.

NABD IS NOT

early warning. Seismometers and existing alert systems own quake detection; we do not compete with them.

NABD IS

the **impact map in the first minute**, and **triage across the first 72 hours** — who is cut off, and where.

Silence is a sensor.

When a whole block of cells goes dark at the same moment, the network already knows where the disaster hit — before a single emergency call gets through.

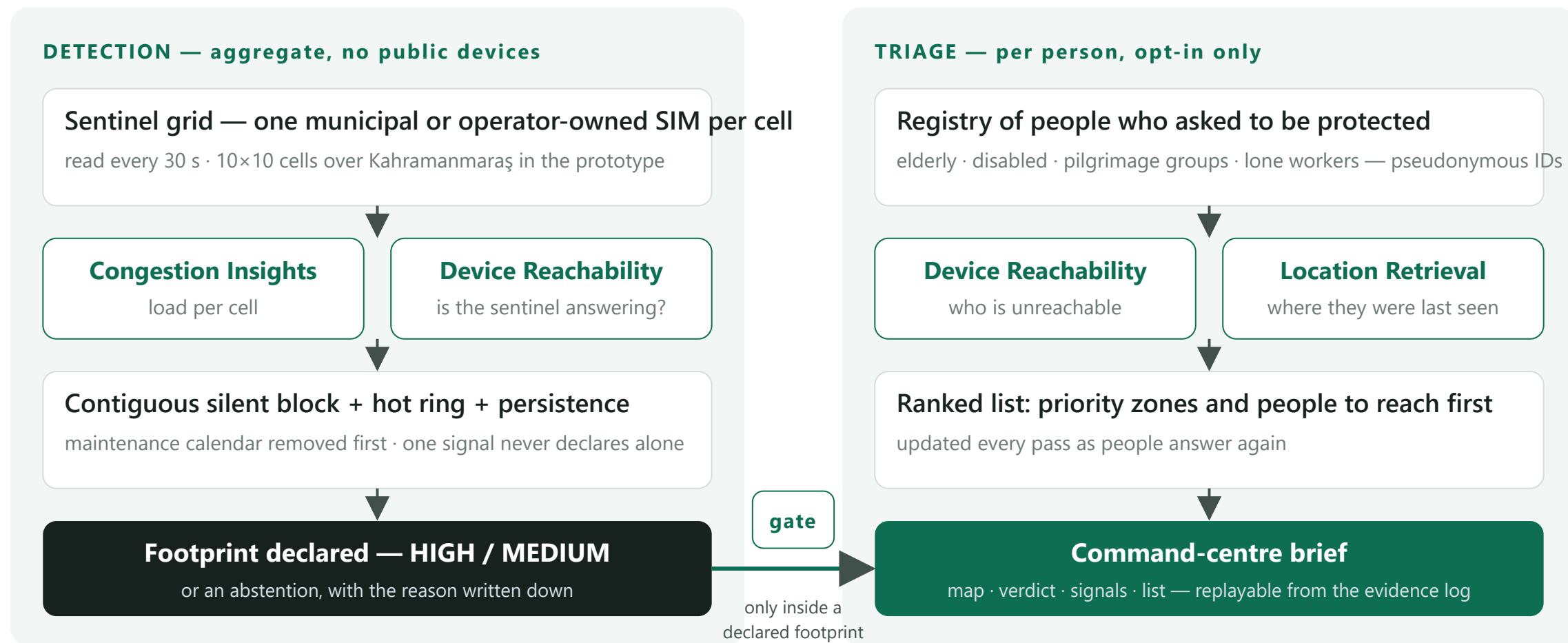
1 An area goes **dark**: its cells stop answering.

2 The ring around it goes **hot**: everyone calls at once.

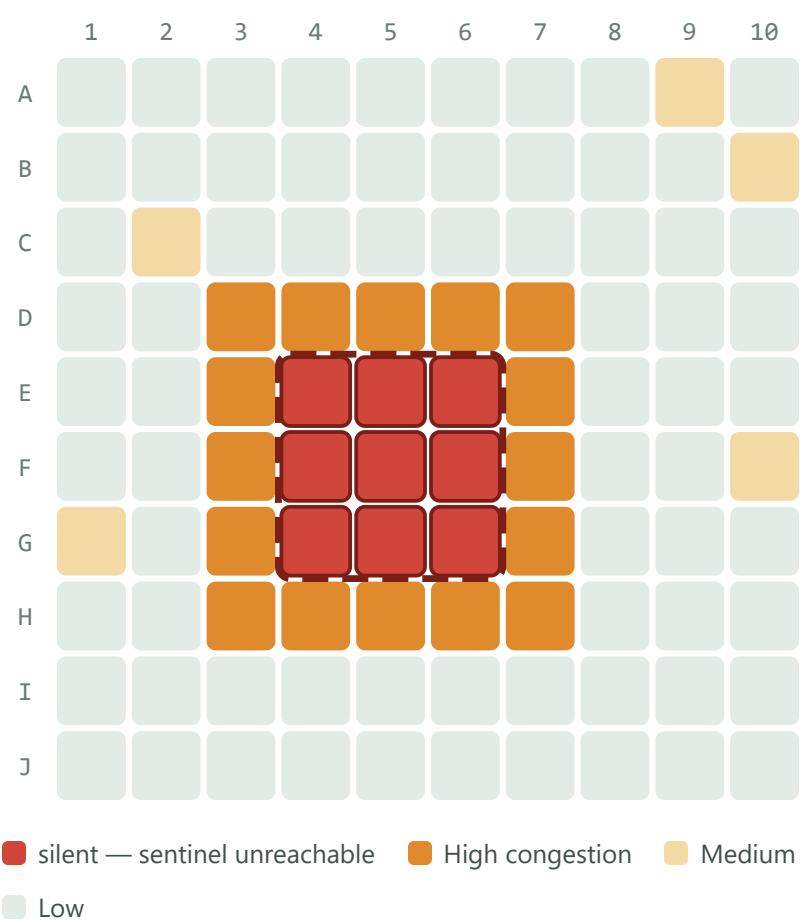
3 That pair, at the same instant, is a signature **no other event produces**.

Disaster-agnostic: earthquake, flood, storm, mass power outage — to the network they are the same thing, an area going silent.

Two layers. Detection is aggregate; triage is consented.



A block goes silent. The ring around it goes hot.



9 cells silent

E4 → G6, about 4.0 km². Their sentinels stop answering — *Device Reachability*.

16 / 16 ring cells at High

Everyone around the block is calling at once — *Congestion Insights*.

Onset synchronised

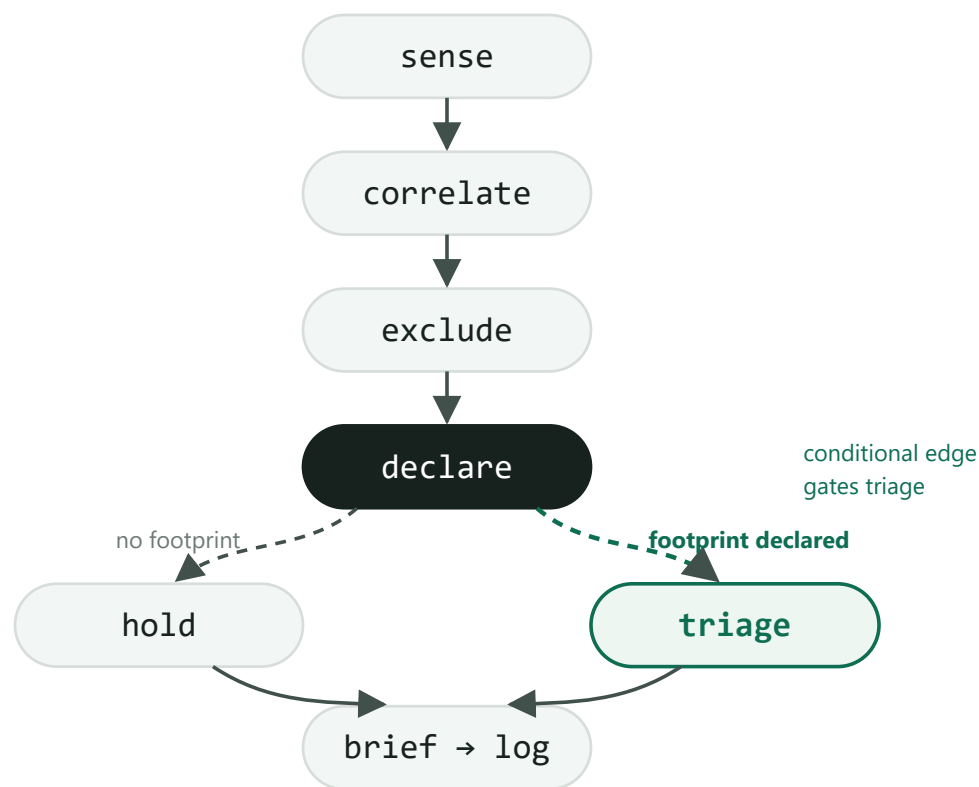
All nine went dark inside one 30-second window. A cell fault does not do that.

Persisted a second pass

Candidate at 09:02:30, declared at 09:03:00 — **55 seconds after onset**, confidence **HIGH**.

The grid is drawn from the evidence record of the declaring pass, not illustrated.

One signal never declares alone.



Two gates, then corroboration. A footprint needs a contiguous silent block *and* silence that is abnormal *there*, measured against those cells' own history. A gate can veto a declaration; it can never cause one.

Grade the confidence. Corroborations — synchronised onset, hot ring — are a list the arithmetic counts, not a hard-coded pair. Two → HIGH. One → MEDIUM. None → abstain, with the reason written into the log. A new network surface is an append to that list, not a redesign.

Gate the personal layer. The conditional edge means no registry device is ever queried outside a declared footprint — a fuzzed test invariant, not a promise.

Keep the LLM off the control path. Detection is deterministic and replayable; the language model only writes the command-centre sentence.

A working agent, offline and live through one surface.

world · model	10×10 grid, 48-person opt-in registry, quake / cell-fault / peak / chronic-flap events, maintenance calendar
gateway	one CAMARA surface, three backends — simulator, live via the Nokia SDK, and replay of a recorded transcript
detector	correlate → exclude → declare; gates and corroborations as named signals, pure functions, no framework import
triage · brief	reachability and last-seen inside the footprint only; the command-centre sentence
graph · loop	the LangGraph runner and the plain runner — same evidence, byte for byte
shakemap · data	the USGS intensity field for 6 Feb 2023, reduced and committed with its provenance
scene · console · parity	five scenes; a self-contained console generated from the evidence; the backend-parity harness

Nabd · نبض

Runs on bare Python

no account, no server, no CDN — the demo cannot fail in the room

Same agent, live

`--backend live` makes the identical three calls through the Nokia SDK and records the raw exchanges

One agent, proven

`python -m nabd.parity` — the agent stack imports no SDK, one function picks the network, every scene replays through the live parsers identically

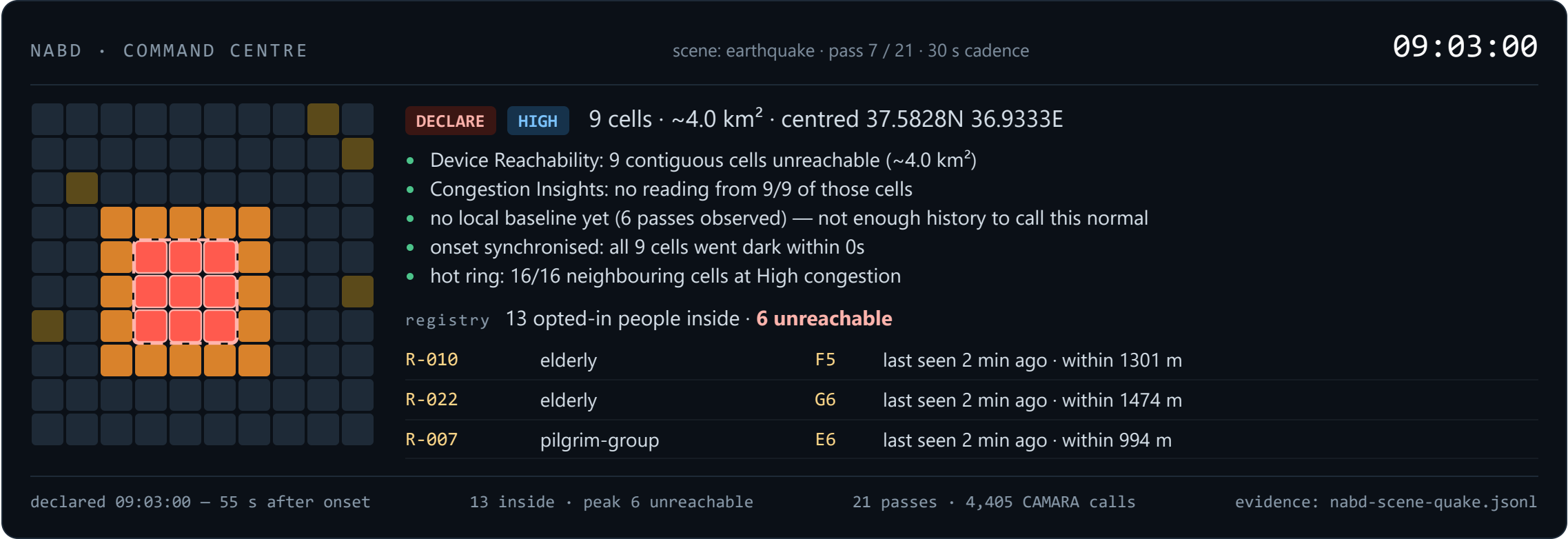
Checked against the real event

the 6 Feb 2023 ShakeMap drives one scene; the declared footprint must stay inside the collapse band the measured shaking defines

83 / 83 tests, 10 suites

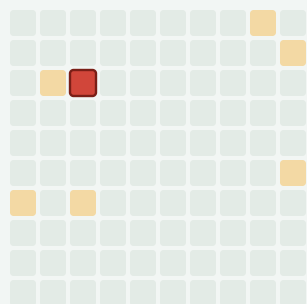
incl. fuzzed invariants: no footprint without a contiguous block; no personal-device call outside a footprint

Declared 55 seconds after onset. Six people to reach first.



Later passes: footprint held; the list updates 6 → 4 as people answer again. Every element on the console is redrawn from the evidence log — the screen is a view over the record, never a second source of truth.

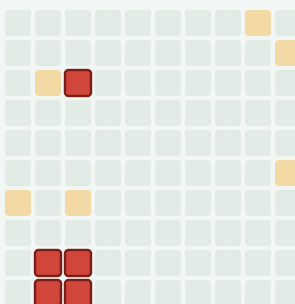
The agent does not cry wolf.



09:01:00 · SINGLE CELL

“single-cell silence at C3: one sentinel unreachable, neighbours normal — consistent with a cell fault, not an impact”

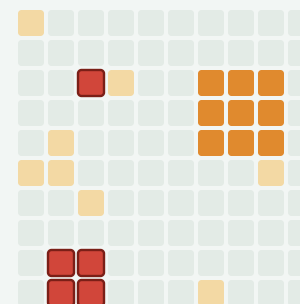
ABSTAIN



09:03:00 · MAINTENANCE

“4 contiguous cells silent, but the block matches maintenance ticket MNT-2214 — expected silence, no alert”

ABSTAIN



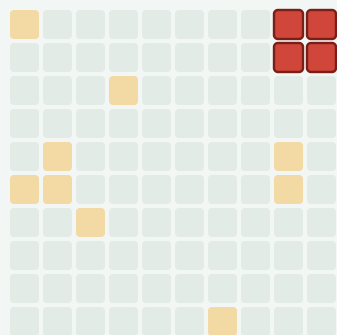
09:05:00 · STADIUM KICK-OFF

“congestion without silence: 9 contiguous cells at High but every sentinel answers — a crowd, not an impact”

ABSTAIN

Noise scene: 21 passes, three abstentions, **zero declarations** — each refusal with its reason in the log. The quiet scene stays quiet. Written reasons are what make a false alarm auditable and a true alarm trusted.

Where silence is normal, silence is not news.

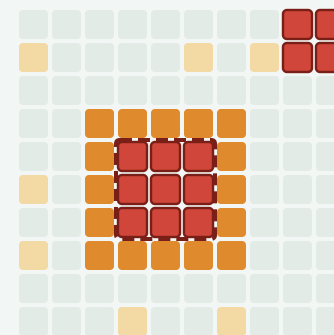


09:05:30 · A BLOCK ON A FAILING BACKHAUL

Four contiguous cells, dark together, no maintenance ticket — **every signal an impact has**. The first three flaps are held for confirmation and come back before it is due. By the fourth, the agent has measured them:

“silence is the local baseline: these 4 cells were already unreachable in 3 of the last 11 passes (27% of the time) — no alert”

ABSTAIN



09:10:30 · SAME RUN, SAME GRID

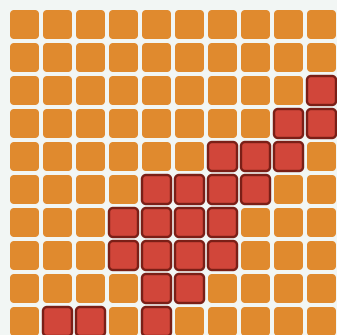
An earthquake hits the centre while the north-eastern block is **still dark**. Two silent blocks on one screen. One alert.

“impact footprint declared: 9 contiguous cells, ~4.0 km²” — HIGH, 30 s after onset, 13 registered people inside

DECLARE

The real question is not whether an outage can be seen — it is whether a **disaster footprint can be told apart from every other reason cells go quiet**. The refusal is measured, not assumed; the declaration in the same run is what shows it is not blindness.

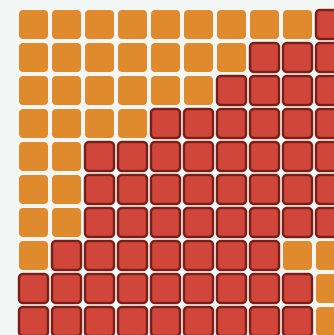
Then we stopped drawing the disaster.



+55 S · THE FIRST MAP

21 contiguous cells, ~2,100 km², HIGH. Every cell it names is one the measured shaking places above the collapse threshold — no invented damage. Two more, an isolated pocket below the three-cell floor, are **deliberately not claimed**.

DECLARE



+9 MIN · THE NETWORK KEEPS DYING

63 cells, ~6,300 km², still HIGH. Masts that survived the shaking lost mains power and drained their batteries, worst-shaken first. The pocket has joined. 22 registered people unreachable, medical-dependent first.

UPDATE x12

The geometry and the intensity in every cell are the **USGS ShakeMap for the M7.8 Pazarcık earthquake**, 6 Feb 2023 — 262 seismic stations, 1,459 intensity observations. Ours is only the rule turning shaking into silence, and both mechanisms are from the field reports; the thresholds are calibrated to Turkcell's “**more than half of local base stations inoperative**” — our window ends 63% dark.

Remove the network APIs and nothing works. The network is the sensor.

API	ROLE	PROTOTYPE
Congestion Insights	detection — load per sentinel cell, the hot ring	live path
Device Reachability Status	detection (sentinels) and triage (registry)	live path
Location Retrieval	triage — last-seen position of the unreachable	live path
Geofencing Subscriptions	boundary events as the footprint grows or shrinks	roadmap
Quality on Demand	a priority channel for responders inside the footprint	roadmap

WHY CAMARA, NOT A CONTROL ROOM

An operator's NOC sees its own cells go down — from inside one operator.

Responders — civil defence, the Red Crescent, municipalities, hospitals — sit **outside**, and cannot log into three operators in the middle of a disaster.

Open Gateway is the **one standard door**: operator-independent, and it unifies every national operator behind a single interface. Stable since 2025, available on Network-as-Code today.

Live where it can be. Simulated where it must be. Private by design.

LIVE VS SIMULATED — AND THE SEAM BETWEEN THEM

The sandbox cannot stage a disaster. So the two claims are separated: the **live platform proves the integration** with real recorded calls; the **simulator proves the scenario**, shaped on the public record of the 2023 Kahramanmaraş outage.

The risk is not the simulator — it is a suspected discontinuity between the paths. So `nabd.parity` measures it: the agent stack imports no SDK and names no backend; **one function** picks the network; and every scene **is recorded and replayed through the live response**

PRIVACY

Detection reads only **sentinel devices the city or operator owns** — never the public.

Its output is **area-level**: cells and a footprint, not people. No cameras, no message content, no tracking.

CONSENT

Triage touches an individual only if they **opted in** to be protected.

The agency holds the registry; the operator honours it. IDs are pseudonymous, and **no location call is ever made outside a declared footprint**.

One engine, many hazards. Minutes, not hours.

IMPACT

Earthquake, flood, storm, mass outage — to the network each is an area going dark, so one engine covers them all.

The first dispatch decision moves from **hours to minutes**, and the most vulnerable are reached first.

SCALE

A sentinel grid scales by **cell count**; SIMs are municipal or operator-owned, so a national rollout is a procurement, not a behaviour change.

Open Gateway makes the same agent portable across operators and countries in the region.

BUSINESS

Buyers: civil-defence agencies (AFAD), Red Crescent societies, municipalities, hospital networks.

Seller: the operator, as an Open Gateway **impact-feed product** — a standing subscription for the grid, per-event triage on top.

ROADMAP Population Density Data as the detection upgrade once published on NaC · Geofencing for boundary events · QoD priority for responders inside the footprint · multi-operator fusion

The network already knows. Nabd makes it say so — in the first minute.

NEXT

Live contract run on Network-as-Code devices ·
pilot with one civil-defence agency on one city's
sentinel grid · Population Density upgrade when
published

TEAM

Kadir Can Yıldırım — Kadir's Team, solo builder,
Türkiye. Code, evidence logs and the command-
centre console are in the submission links.

See you in Doha — MWC Doha, November 2026.

